

Recap

DNS database consists of billions of Resource Records that provide the mappings between names and IP addresses of all devices connected to the Internet

Each Name Server in the hierarchy stores a (small) subset of the entire database

Multiple Name Servers associated with each node of the DNS tree to ensure the availability of the service

Resource Records are characterized by a simple (standard) structure

Various types of records

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RR types: basic

Type **A** (Address) or **AAAA**: specifies the IP address(es) corresponding to a given hostname [*forward mapping*]

Example: NAME: FQDN `eva.unipv.it`
 RDATA: IP address `193.204.34.2`
 [*one-to-one mapping*]

In the *one-to-many* mappings, all IP addresses are listed in the data field

Type **MX** (Mail eXchanger): specifies the mail server(s) associated with a domain

Example: NAME: domain name `universitadipavia.it`
 RDATA : name of the `ALT2.ASPMX.L.GOOGLE.COM`
 `ASPMX.L.GOOGLE.COM`
 `ASPMX4.GOOGLEMAIL.COM`
 `ASPMX2.GOOGLEMAIL.COM`
 `ASPMX3.GOOGLEMAIL.COM`
 `ASPMX5.GOOGLEMAIL.COM`
 `ALT1.ASPMX.L.GOOGLE.COM`

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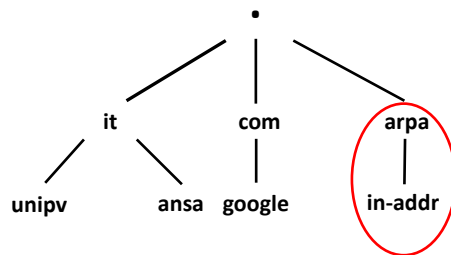
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RR types: basic (cont'd)

Type PTR (Pointer): maps an IP address to a hostname [*reverse mapping*]

Example: NAME: reverselPaddress.in-addr.arpa
3.34.204.193.in-addr.arpa
RDATA: FQDN poldo.unipv.it



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RR types: zone

Type **NS** (Name Server): defines the names of the Authoritative Name Servers of a domain [infrastructure record]

Example:	NAME: domain name	unipv.it
	RDATA: names of the Name Servers	ipv36.unipv.it ipv512.unipv.it

Type **SOA** (Start Of Authority): mandatory record used to specify the core characteristics and attributes of a domain [first record of the zone file]

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SOA Resource Record

SOA data:

- *origin*: name of the primary Name Server of the domain
- *mail addr*: email address of the domain administrator
- *serial*: serial number (e.g., date of last change & version number)
- *refresh*: time [s] indicating the interval before the zone should be refreshed
- *retry*: time [s] indicating the interval that should elapse before a failed refresh should be retried
- *expire*: time [s] indicating the upper limit of the interval that can elapse before the zone is no longer authoritative
- *TTL*: minimum TTL [s] for the RRs of the zone

NOTIFY mechanism reduces the need for frequent checks of the SOA record [RFC 1996 "A Mechanism for Prompt Notification of Zone Changes (DNS NOTIFY)"]

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SOA Records

root-servers.net

```
primary name server = a.root-servers.net
responsible mail addr = nstld.verisign-grs.com
serial = 2026030500
refresh = 14400 (4 hours)
retry = 7200 (2 hours)
expire = 1209600 (14 days)
default TTL = 3600000 (41 days 16 hours)
```

net

```
primary name server = a.gtld-servers.net
responsible mail addr = nstld.verisign-grs.com
serial = 1774559662
refresh = 1800 (30 mins)
retry = 900 (15 mins)
expire = 604800 (7 days)
default TTL = 900 (15 mins)
```

it

```
primary name server = dns.nic.it
responsible mail addr = hostmaster.nic.it
serial = 2026032522
refresh = 10800 (3 hours)
retry = 900 (15 mins)
expire = 604800 (7 days)
default TTL = 3600 (1 hour)
```

ai

```
primary name server = v0n0.nic.ai
responsible mail addr = hostmaster.donuts.email
serial = 1774473321
refresh = 7200 (2 hours)
retry = 900 (15 mins)
expire = 1209600 (14 days)
default TTL = 3600 (1 hour)
```

io

```
primary name server = a0.nic.io
responsible mail addr = hostmaster.donuts.email
serial = 1774558819
refresh = 7200 (2 hours)
retry = 900 (15 mins)
expire = 1209600 (14 days)
default TTL = 3600 (1 hour)
```

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Second Level Domain SOA Records

unipv.it primary name server = ipv36.unipv.it responsible mail addr = root.ipv36.unipv.it serial = 2026032603 refresh = 12000 (3 hours 20 mins) retry = 1200 (20 mins) expire = 1209600 (14 days) default TTL = 300 (5 mins)	ucsd.edu primary name server = act-vnios-2.ucsd.edu responsible mail addr = hostmaster.ucsd.edu serial = 1462920710 refresh = 10800 (3 hours) retry = 3600 (1 hour) expire = 2419200 (28 days) default TTL = 900 (15 mins)
google.com primary name server = ns1.google.com responsible mail addr = dns-admin.google.com serial = 889730864 refresh = 900 (15 mins) retry = 900 (15 mins) expire = 1800 (30 mins) default TTL = 60 (1 min)	blog.google primary name server = ns1.googledomains.com responsible mail addr = dns-admin.google.com serial = 2017070401 refresh = 21600 (6 hours) retry = 3600 (1 hour) expire = 1209600 (14 days) default TTL = 300 (5 mins)
twitch.tv primary name server = ns-219.awsdns-27.com responsible mail addr = awsdns-hostmaster.amazon.com serial = 1 refresh = 300 (5 mins) retry = 150 (2 mins 30 secs) expire = 604800 (7 days) default TTL = 60 (1 min)	tiktok.com primary name server = a9-66.akam.net responsible mail addr = hostmaster.akamai.com serial = 1554957832 refresh = 43200 (12 hours) retry = 7200 (2 hours) expire = 604800 (7 days) default TTL = 7200 (2 hours)

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RR types: security

These record types refer to DNS security extensions, that is, a collection of Resource Records and protocol modifications providing source authentication for the DNS [RFC 4034 “Resource Records for the DNS Security Extensions” published in 2005]

Type DNSKEY : specifies the public key associated with a domain

Example: NAME: domain name
 RDATA: public key

Type RRSIG : specifies the digital signature for a set of Resource Records

Type NSEC (Next Secure): specifies the types of Resource Records associated with a domain and the next authoritative name after the domain (used for *authenticated denial of existence*)

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RR types: optional

These record types refer to *optional* features supported by DNS

Type **CNAME** (Canonical Name): specifies the canonical name corresponding to an alias (e.g., CDN)

Example: NAME: alias `www.ucsd.edu.`
RDATA: canonical name `cms-web-1573499122.us-west-2.elb.amazonaws.com.`

Type **TEXT** (Text): used to associate text with a domain name to be used for different objectives

Example: NAME: domain name `unipv.it.`
RDATA: textual content

```
"atlassian-domain-verification=S2XG48dLZRERXgcf6tystt9yM  
"google-site-verification=ZY0rml0atjmdx-kU-yATJatNbSGVQ  
"HARICA-y2rgha3HYUmhgjTEwj2"  
"docuSign=7015b115-44f9-40af-8552-7bf780f2b533"  
"amazonSES:HNSFW27jQcspFg0seyCD2g2v3SwqqG7iYzWxVLIizw8=  
"facebook-domain-verification=8hfo9h1dtvrmuhvv3ptr8g7884  
"MS=421AA9E190F698E46727F29DA1D625BF605AEF88"
```

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Other RR types

The service record (**SRV**) is used to discover the services offered by a domain [RFC 2782 "A DNS RR for specifying the location of services (DNS SRV)" published in 2000]

A service is identified by a symbolic name, the transport protocol used and the domain name (i.e., **_Service._Proto.Name**)

```
_xmpp._tcp.xyz.it. 86400 IN SRV 10 5 5223  
server.xyz.it.
```

SVCB (Service Binding) and **HTTPS** RRs aim at facilitating the lookup of information needed to connect to network services ahead of the connection [RFC 9461 "Service binding and parameters for DNS Servers (SVCB and HTTPS Resource Records)" November 2023]

Benefits for performance and privacy (RTTs and unencrypted messages)

```
abc.com 300 IN HTTPS 1 cdn.abc.net alpn="h3","h2"
```

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Root hints and zone files

The root hints are a set of RRs used to locate Name Servers that are *authoritative* for the root of the DNS tree

Root hints

Each Name Server stores its own *zone file*

A zone file is a text file that contains the RRs referring to the domain associated with the Name Server

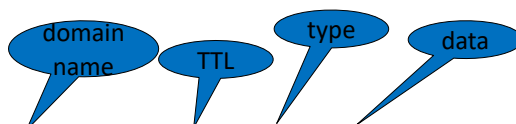
Root zone

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RRs associated with `it.`



```
it. 172800 IN NS a.dns.it.
it. 172800 IN NS m.dns.it.
it. 172800 IN NS r.dns.it.
it. 172800 IN NS v.dns.it.
it. 172800 IN NS dns.nic.it.
it. 172800 IN NS nameserver.cnr.it.
it. 86400 IN DS 41901 10 2 47F7F7BA21E48591F6172EED13E35B66893AD9F2880FC9BADA64F68CE28EBB90
it. 86400 IN RRSIG DS 8 1 86400 20250410050000 20250328040000 26470 .
U6CbFjT2zhwFkQ0t142+w90aYHn0paS1TCuOHCjgqtkwPKOp4j3K5vOCxMBtYFHE2Q+N1IefVx344CylvaY9mAMVxmv4aRLpFxJ37CzaQkTc1xHH1U11T9Hst1+yRHM3gmDRLf58XVjrs5l
EE5jtQsDerA0s4Mo4Sfr9012oiBs0KyGVuABZp1miTjadbH97AE1xCRVpS59Y5R+Z/XSGVxemb/e6gP517KPyok/qZ7c0ZU311YkIWBh5bDu2oDz+Mykyca+
+eEujiTU+RpyTwL1j9EPxs8cddlP1ac3QQCmRvYFEBX14CexgoNduHuFpaS21Wna6054wnw==
it. 86400 IN NSEC itau. NS DS RRSIG NSEC
it. 86400 IN RRSIG NSEC 8 1 86400 20250410050000 20250328040000 26470 . ZQjmJqSwyp8F0hu7DC8Ctq/
Tc8Yiak3mcXCg0gfruielSnR5EuQ02HQHdS05gs87gMfo4MvrIa0z5rDykb73F51svcrDA9J6WlkJGAgDaFPYCYNPFaUoagRUXIZISmv0DEd+G+MhV9bJBVRtON4e/bpQhRzV4pnq896q
QxX3nk2X3fN1HFK24mAcDk801S8pzuTAYjz2e2TgR5anJXG6jri4+2dIN8pqC0o1a+iRq98jevXs60L2TQXmf1BAMp7QIgbtSbHaxkVc1B1S11yIcflfbJNLraJfr/mzco/nI8CDF3zpw1IC
nameserver.cnr.it. 172800 IN A 194.119.192.34
nameserver.cnr.it. 172800 IN AAAA 2a00:1620:c0:220:194:119:192:34
a.dns.it. 172800 IN A 194.0.16.215
a.dns.it. 172800 IN AAAA 2001:678:12:0:194:0:16:215
m.dns.it. 172800 IN A 217.29.76.4
m.dns.it. 172800 IN AAAA 2001:1ac0:0:200:0:a5d1:6004:2
r.dns.it. 172800 IN A 193.206.141.46
r.dns.it. 172800 IN AAAA 2001:760:ffff:ffff:0:0:0:ca
v.dns.it. 172800 IN A 194.0.25.44
v.dns.it. 172800 IN AAAA 2001:678:20:0:0:0:0:44
osiris.nameex.it 172800 IN A 193.201.40.6
osiris.nameex.it 172800 IN AAAA 2a0f:80:1:0:0:0:0:6
seth.nameex.it. 172800 IN A 193.201.40.7
seth.nameex.it. 172800 IN AAAA 2a0f:80:1:0:0:0:0:7
dns.nic.it. 172800 IN A 192.12.192.5
dns.nic.it. 172800 IN AAAA 2a00:d40:1:1:0:0:0:5
itau. 172800 IN NS a.nic.itau.
```

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Examples of RRs for google.

google.	172800	IN	NS	ns-tld1.charlestonroadregistry.com.
google.	172800	IN	NS	ns-tld2.charlestonroadregistry.com.
google.	172800	IN	NS	ns-tld3.charlestonroadregistry.com.
google.	172800	IN	NS	ns-tld4.charlestonroadregistry.com.
google.	172800	IN	NS	ns-tld5.charlestonroadregistry.com.
google.	86400	IN	DS	6125 8 2 88f88780231071535788AD3800E9543500474E5C30C29698B40A3DB23ED9DA9F
google.	86400	IN	RRSIG	DS 8 1 86400 20230410050000 20230328040000 951 .
EKfUqg/AGEnl0tj4LSGgraeb1ggstpb8C6sp+5ZkYLXWGHNCaFoFTPDL58FHCc6pT3CH21J/nY5wKDT9dvbB87xKpGkF5F+9pN+k/so3X1R/iacDd3wxHDBpc				
/GKa7nFlKocK+RQk5P3VDWqZmFzMQh4nxWAcGq5C3wFN+hLRrQQGYUMYHOSYkamcN/123TxCZ1Yf513wFV2				
/iWhWuHfri7bcMGBruPOFwemBgZ287mqubv26nVvZcOp2055w3uHgrTUNjrRC4kXo1rj01uHj0e77wAZHzvg0/Fv1yZMVRfvTFMFsFgnQ5g0+ic0+Lc09xNudpw7gx+QzCLRsCdG				
google.	86400	IN	NSEC	gop. NS DS RRSIG NSEC
google.	86400	IN	RRSIG	NSEC 8 1 86400 20230410050000 20230328040000 951 .
JRDvxRa/biNMiJLetv0P6qNA+GVafG3jws5cVT7ZMJmp0sP+AKzbGmwOyVzmngJESDzh				
/AjYRkpQ19IF39Cd1QXLNCU5N3bxALu65YtEO+mh8QcwD3w5gZcJvGxgf3sMBZd6oV18Rbcgq10D57PUeto54jGo3hbwrzi+fk2Kgle10gfGC3				
/zQroGhXrvNDrTvPI70FenQ2IMQH1QIH1eC				
/w4UBDYEPrsYAdMLzkGhdh6xPMK1B3CoXSU7YTEu5+1jDRhsuemJLNnleUXoFvmwLFF6LK9bt4AAtwIAnIhYf5ak01hmTA3mv2onPMTXZ78bYvrM30EKKYImM28GBas6R==				

Registrant:

Name: Charleston Road Registry Inc.

Organization: Charleston Road Registry Inc.

Mailing Address: 1600 Amphitheatre Parkway | Mountain View CA 94043 United States of America (the)

Administrative:

Name: TLD Admin

Organization: Google Inc.

Email: iana-contact@google.com

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Database updates

At least two Name Servers associated with a zone:

- Primary (i.e., master) Name Server
- Secondary (i.e., slave) Name Server(s)

RRs are *manually* entered into a zone file and loaded into the primary (authoritative) Name Server for the zone

Each secondary NS could poll the primary NS or it might receive automatic notifications from the primary NS (e.g., **NOTIFY** option)

Changes to the master copy of the database are propagated to the secondary NSs according to timings (e.g., refresh parameter) set by the domain administrator in the SOA RR (loose coherency)

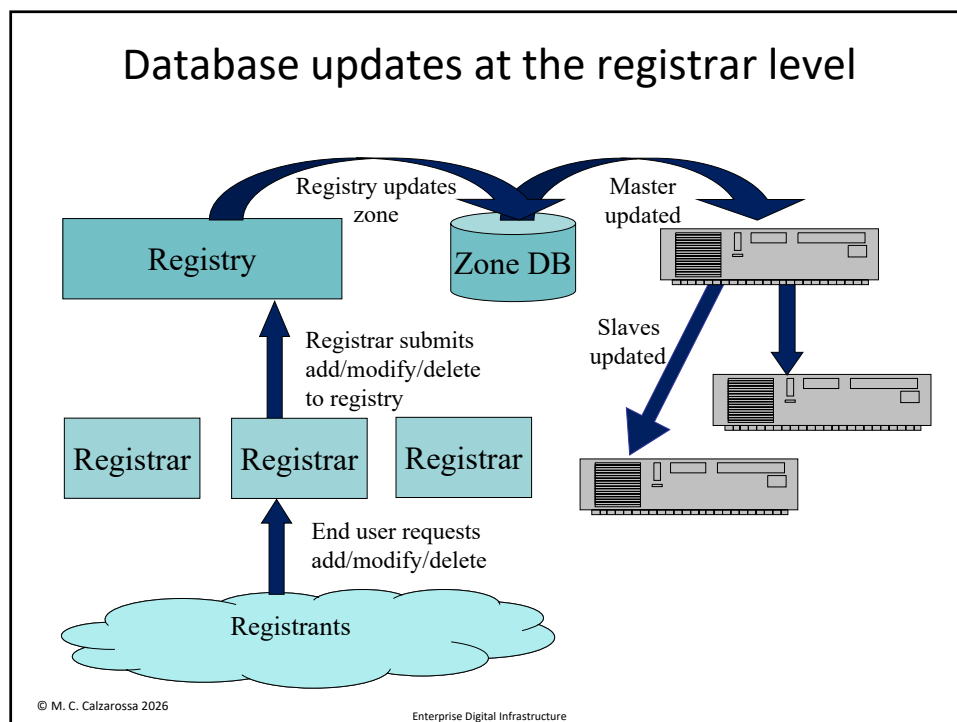
A serial number is associated with each version of a zone file

This number is modified whenever the administrator changes anything

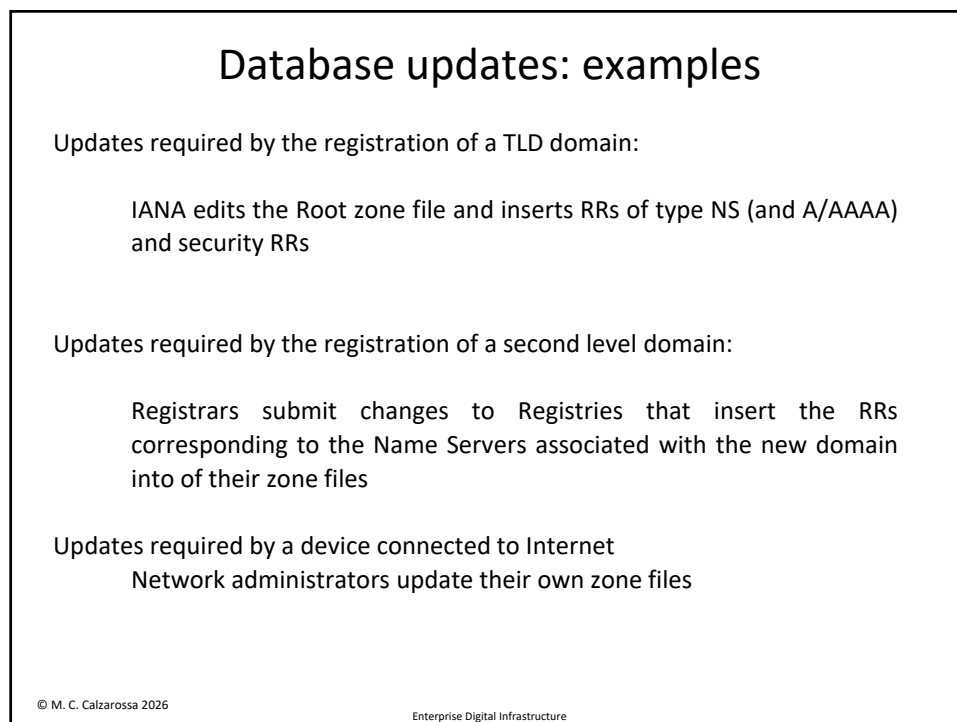
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DNS setup: user perspective

How do we associate a Name Server with a device?

It depends on the policy adopted by the ISP on the usage of IP addresses

1. Static IP address
2. Dynamic IP address assigned via DHCP

In case of a dynamic IP address, DHCP *automatically* assigns the IP address of the Name Server(s) but we can also set them *manually*

In case of a static IP address, it is necessary to specify the IP addresses of the Name Servers of the domain or of “open” Name Server(s) (e.g., **/etc/systemd/resolved.conf** under Linux)

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DNS

Two main components:

- ✓ 1. a *distributed database* implemented on many *geographically distributed servers* characterized by a *hierarchical organization*
- 2. an application layer *protocol* based on a client/server model over the UDP protocol (on port 53) used to query the database (i.e., to provide the translation service)

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DNS protocol overview

DNS is a *lightweight application layer protocol* used to query the database

Client/server model

Stub resolver is the client component that runs on the user device and is built in the operating system

Full resolver is a client and server component that runs on Name Servers

Two types of messages exchanged by clients and servers

- query messages sent by clients
- response messages sent by servers

UDP protocol for efficiency (reliability not ensured) and port 53

Implementations based on TCP mainly used for maintenance/update purposes or to ensure security/privacy

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DNS protocol: example

`ssh eva.unipv.it`
hostname used by stub resolver

lookup phase

The stub resolver

1. creates the DNS query from the request received by the application for translating the hostname into the corresponding IP address
2. sends the query to a full resolver
3. receives the answer and returns it to the application

Two possible outcomes of a query:

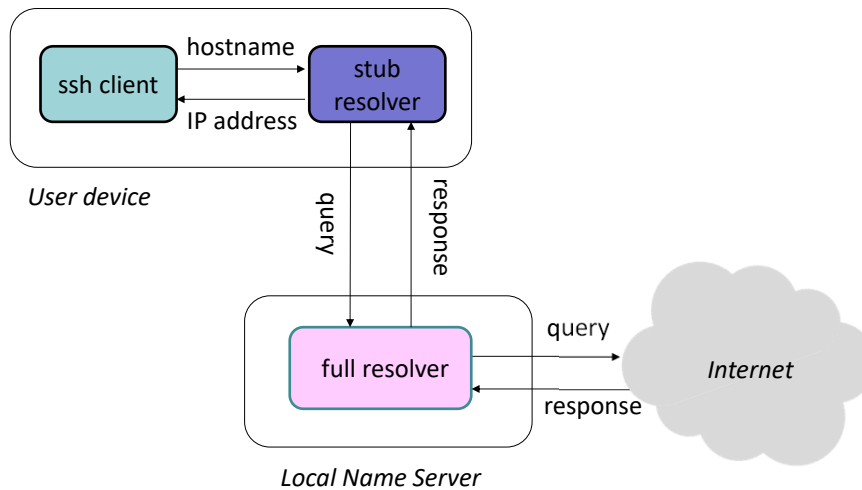
- *success*: the answer provides the IP address corresponding to the name specified by the query
- *failure*: the answer contains an error (e.g., **NXDOMAIN**)

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Which Name Server(s)



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Iterated vs recursive queries

An *iterated* query is centered around the local Name Server that is responsible to send queries and collect answers

Each NS can provide

- a *complete answer*
- a *partial answer*
- an *error* (due to a permanent or temporary failure)

A *recursive* query puts the responsibility of the answer on the specific Name Server being contacted

Each NS can provide

- a *complete answer*
- an *error* (due to a permanent or temporary failure)

The queries between clients and local Name Servers are always recursive, while Root and TLD Name Servers support iterated queries only

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Example of iterated query

Time spent for the lookup phase: **4**
RTTs (1 "local" & 3 "remote")

